

Predictive Forecasting of Care Load & Placement Demand

HHS Unaccompanied Alien Children (UAC) Program — Detailed Research Paper

Abstract

This study develops a short-term forecasting and decision-support framework for HHS care load and placement demand using 720 valid observations from 2023-01-12 to 2025-12-21. The workflow combines data validation, exploratory analysis, feature engineering, chronological model evaluation, uncertainty estimation and historical capacity-stress monitoring. Four forecasting approaches were tested: Naive persistence, 7-observation moving average, ARIMA(1,1,1), and Random Forest. The Naive model produced the best held-out performance with MAE 10.24, RMSE 14.05 and MAPE 0.454%. The findings show that recent HHS care load is highly persistent and that a simple baseline can outperform more complex methods for this evaluation period.

1. Introduction

Care-load forecasting is a time-dependent planning problem. The number of children in HHS care changes as children enter through transfers and leave through discharges. A reliable short-term forecast can help analysts monitor likely care-load direction, identify periods requiring closer attention, and compare forecast uncertainty with historical conditions.

2. Objectives

The project has five objectives: forecast Children in HHS Care over short horizons; estimate discharge demand; quantify transfer-versus-discharge pressure; compare statistical and machine-learning forecasting approaches using unseen data; and create a historical early-warning indicator for unusually high care load.

3. Dataset

The cleaned dataset contains 720 observations and six original variables: date, children apprehended and placed in CBP custody, children in CBP custody, children transferred out of CBP custody, children in HHS Care, and children discharged from HHS Care. The observation period is 2023-01-12 to 2025-12-21. The primary target is Children in HHS Care.

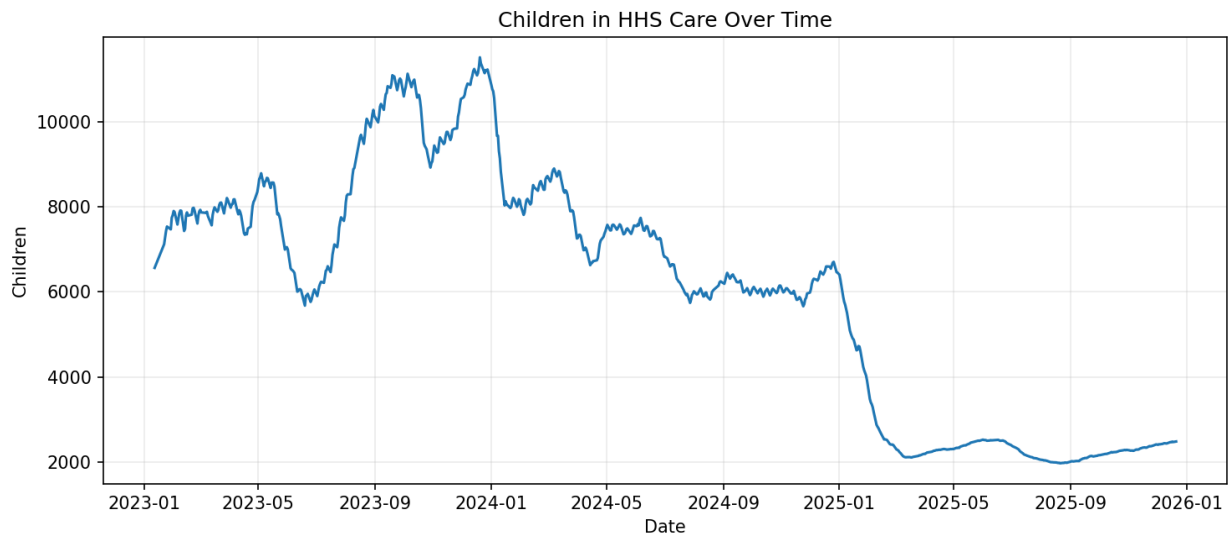
4. Data Cleaning and Validation

Completely blank rows were removed. Dates were converted to datetime and sorted chronologically. Numeric fields were converted to numeric values, including removal of comma separators in HHS care values. Duplicate-date and missing-value checks were performed. The date span contains 1075 calendar days but only 720 observed dates, leaving 355 unobserved calendar dates. Because flows cannot safely be invented for missing reporting dates, the analysis does not interpolate the missing observations.

5. Exploratory Data Analysis

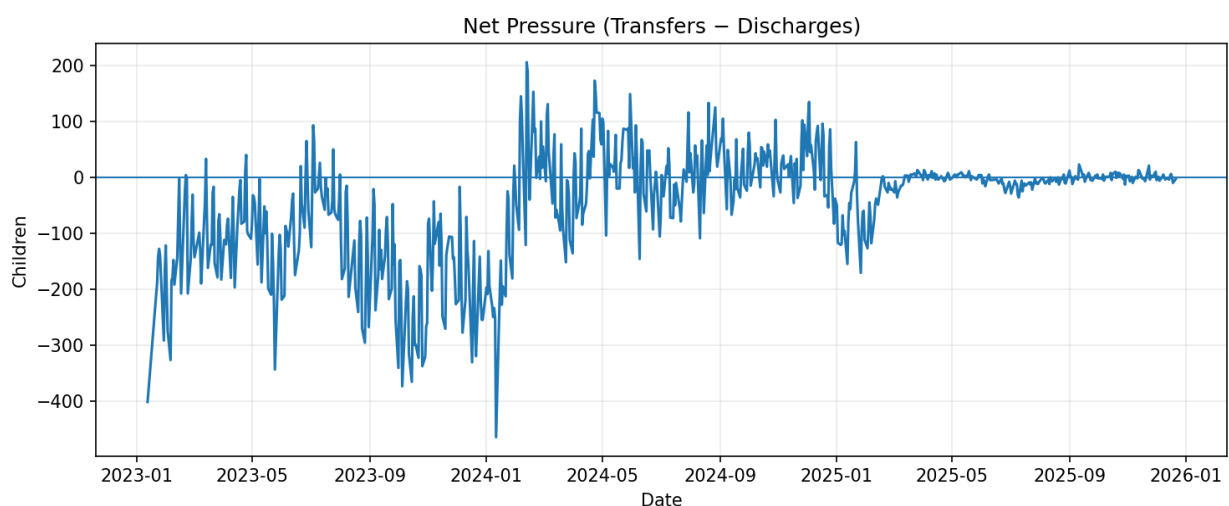
5.1 HHS Care Load Trend

The series exhibits several distinct regimes. HHS care was around the mid-thousands to 8,000 range during early 2023, rose strongly during the second half of 2023, and reached a maximum of **11,516** children on 2023-12-20. It then declined through 2024 and fell to a minimum of **1,972** on 2025-08-21. By the end of the sample, the load had recovered to **2,484**. The presence of these large structural changes is important when interpreting model performance.



5.2 Transfer, Discharge and Net Pressure

Net pressure is defined as transfers into HHS minus discharges from HHS. Its mean is -44.74. Negative pressure occurred 475 times, positive pressure 238 times, and zero pressure 7 times. Negative pressure was therefore more frequent, particularly during periods when the overall HHS care load was declining.



5.3 Correlation

The correlation analysis showed strong relationships among operational variables. Children in HHS Care correlated 0.71 with transfers out of CBP custody and 0.92 with HHS discharges. Net pressure correlated -0.48 with HHS care and -0.64 with discharges. These are associations rather than causal effects, but they confirm that the variables contain substantial shared operational information.

6. Feature Engineering

The forecasting dataset uses HHS care lags at 1, 7 and 14 observations; 7- and 14-observation moving averages; one-observation transfer and discharge lags; and Gap_Days, the number of calendar days since the previous report. Observation-based lags are used because the source is irregularly reported. Thus, a 7-observation lag is not necessarily seven calendar days.

7. Forecasting Methodology

A strict chronological split was used to avoid future-data leakage. After lag creation, 691 usable observations were available: 552 training observations and 139 testing observations. Training covered 2023-03-07 through 2025-05-21; testing covered 2025-05-22 through 2025-12-21. Models were evaluated using MAE, RMSE and MAPE.

8. Models

Naive: predicts the next observation using the latest observed care load. **Moving Average:** predicts from the recent 7-observation local level. **ARIMA(1,1,1):** models differenced autocorrelation and moving-average dynamics. **Random Forest:** learns nonlinear relationships from lag, moving-average and reporting-gap features.

9. Model Results

Model	MAE	RMSE	MAPE (%)	Result
Naive	10.24	14.05	0.454	Selected
Moving Average (7)	24.65	30.52	1.101	
Random Forest	54.19	79.18	2.558	
ARIMA (1,1,1)	235.21	285.63	11.114	

The Naive model is the clear winner on the held-out test set. Its MAE of 10.24 means the average absolute error was approximately 10 children, while its 0.454% MAPE indicates very small relative error over the test period. Moving Average was the second-best model. Random Forest was less accurate despite using additional features. ARIMA performed substantially worse and generated a nearly flat forecast around the end-of-training level.

10. Random Forest Feature Importance

The Random Forest analysis showed HHS_Lag_1 as the dominant feature, followed by HHS_MA_7. HHS_Lag_7, HHS_MA_14 and HHS_Lag_14 contributed much less, while transfer lag, discharge lag and reporting gap had near-zero importance. This supports the central finding that recent HHS care level is the strongest short-term signal.

11. Final Forecast

The final forecasting baseline uses the latest observed HHS care load of **2,484** children. For 7-, 14- and 30-observation horizons, the Naive point forecast remains 2,484. This should be interpreted as a persistence forecast rather than a claim that the population will remain exactly constant on every future calendar day.

Horizon	Point forecast	Use
7 observations	2,484	Near-term planning
14 observations	2,484	Short-term planning
30 observations	2,484	Extended short-term view

12. Forecast Uncertainty

Approximate 95% prediction intervals were estimated from the historical variability of one-step changes, with uncertainty increasing with the square root of the horizon. This is a practical communication device rather than a formally calibrated probabilistic model. A production implementation should calibrate intervals using walk-forward residuals.

13. Discharge Demand

The discharge module uses the same persistence principle for Children discharged from HHS Care. The latest observed discharge value is 14. Approximate 95% ranges are 0–304 for 7 observations, 0–425 for 14 observations, and 0–615 for 30 observations. The much wider discharge ranges indicate that flow demand is substantially more volatile than the HHS stock measure.

14. Capacity Stress Indicator

No official capacity limit is supplied. Therefore, the project uses the historical 90th percentile as an analytical threshold: **9,762** children. The latest load of 2,484 is below this threshold and is classified as LOW under the historical indicator. This must not be interpreted as an official HHS capacity assessment.

15. Key Findings

- The HHS care-load series contains large structural changes, including a late-2023 peak and a major 2024–2025 decline.
- Short-term care load is highly persistent.
- Negative net pressure is more common than positive pressure.
- The Naive model outperforms all tested alternatives on the held-out period.
- Random Forest confirms the importance of the most recent HHS care value.
- Discharge demand is more volatile and therefore carries wider uncertainty.
- The latest care load is low relative to the historical 90th-percentile threshold.

16. Recommendations

1. Use the Naive model as the current benchmark for short-term care-load forecasting. 2. Monitor transfer-minus-discharge pressure alongside the forecast. 3. Display uncertainty bands rather than only point estimates. 4. Re-evaluate the model with walk-forward validation as new observations arrive. 5. Preserve reporting gaps rather than inventing flow values. 6. Replace the historical stress threshold with an authoritative capacity measure when available.

17. Limitations and Future Work

The major limitation is irregular reporting, which makes calendar-day interpretation different from observation-based forecasting. The capacity threshold is analytical rather than official. The dataset lacks external drivers such as policy changes, staffing, facility capacity and other contextual variables. Future work should use calendar-aware models, rolling-origin validation, calibrated probabilistic intervals and external operational covariates.

18. Conclusion

The project demonstrates that a simple, transparent persistence model can provide excellent short-term forecasting performance for this dataset. With MAE 10.24 and MAPE 0.454%, Naive forecasting is the most defensible selected model under the current evaluation design. Combined with flow-pressure analysis, discharge uncertainty and historical stress monitoring, the resulting framework provides a practical foundation for decision support.